

1 Refresher

1. Prove $E[(x_i - \mu)x_i] = \sigma^2$ and $E[x^2] - E[x]^2 = \sigma^2$:

(a) By definition of variance, $\sigma^2 = E[(x_i - \mu)(x_i - \mu)] = E[(x_i - \mu)x_i] - E[(x_i - \mu)\mu]$. Note $E[(x_i - \mu)\mu] = \mu \times E[(x_i - \mu)] = 0$ as μ is just a constant and $E[x_i - \mu] = 0$ by definition of μ as the population mean.

(b) $\sigma^2 = E[(x_i - \mu)^2] = E[x_i^2 - 2x_i\mu + \mu^2] = E[x_i^2] - 2E[x_i]\mu + \mu^2 = E[x_i^2] - \mu^2$.

2. Show sample variance $s_n = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$ is an unbiased estimator for σ^2 .

$$\begin{aligned} E[s_n] &= E\left[\frac{1}{n-1} \sum_i (x_i - \bar{x})^2\right] \\ &= \frac{1}{n-1} E\left[\sum_i (x_i - \bar{x})^2\right] \\ &= \frac{n}{n-1} E[(x_i - \bar{x})^2] \text{ using the iid assumption} \\ &= \frac{n}{n-1} E[(x_i - \mu + \mu - \bar{x})^2] \\ &= \frac{n}{n-1} (E[(x_i - \mu)^2] + E[(\bar{x} - \mu)^2] - 2E[(x_i - \mu)(\bar{x} - \mu)]) \end{aligned} \quad (1)$$

Now let's break down each term in (1): $E[(x_i - \mu)^2] = \sigma^2$ by definition. The sample mean $\bar{x} = \frac{1}{n} \sum_i x_i$ has population mean $E[\bar{x}] = E[x_i] = \mu$, and therefore

$$E[(\bar{x} - \mu)^2] = \text{var}(\bar{x}) = \sigma^2/n.$$

By iid assumption ($\{x_i : i = 1 \dots n\}$ is a random sample where x_i 's are independently drawn from the same distribution),

$$\begin{aligned} E[x_i x_j] &= 0 \text{ if } i \neq j \\ \Rightarrow E[(x_i - \mu)(\bar{x} - \mu)] &= E[(x_i - \mu)\left(\frac{x_i}{n} - \mu\right)] \\ &= \frac{1}{n} E[(x_i - \mu)x_i] - \underbrace{\mu E[x_i - \mu]}_{=0} \\ &= \frac{1}{n} \sigma^2 \end{aligned}$$

So, back to (1):

$$\begin{aligned} E[s_n] &= \frac{n}{n-1} (E[(x_i - \mu)^2] + E[(\bar{x} - \mu)^2] + 2E[(x_i - \mu)(\bar{x} - \mu)]) \\ &= \frac{n}{n-1} \left(\sigma^2 + \frac{1}{n} \sigma^2 - \frac{2}{n} \sigma^2 \right) = \sigma^2 \end{aligned}$$

3. Regression x_i on a constant: $x_i = \beta * 1 + \epsilon_i$:

$$\begin{aligned} \beta &= \text{argmin}_b E[(x_i - b)^2] \\ \text{FOC: } -2E[(x_i - b)] &= 0 \\ \Rightarrow \beta &= E[x_i] \text{ solves the FOC} \end{aligned} \quad (2)$$

Note the first order condition - (2) represents $E[\epsilon_i * 1] = 0$, that is, the residual in the population regression is orthogonal the regressor 1.

In a random sample, the OLS estimator:

$$\hat{\beta} = \underset{b}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n (x_i - b)^2$$

$$\text{FOC: } -2 \frac{1}{n} \sum_{i=1}^n (x_i - b) = 0$$

$$\Rightarrow \hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i \text{ solves the FOC}$$

$\hat{\beta}$ is the sample mean of dependent variable x_i .